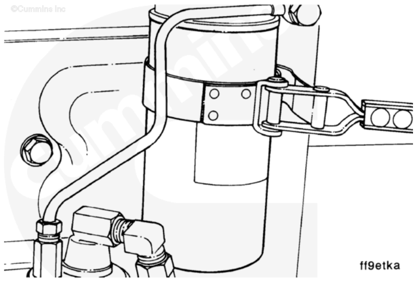
Trainings

Initial Check

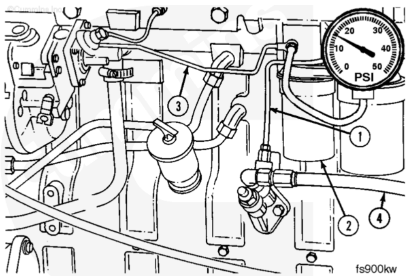
Measure the pressure drop across the fuel filter at low idle. If the filter restriction is above the maximum, it **must** be replaced.

	kpa		psi	
Piston Lift Pump		34	MAX	5



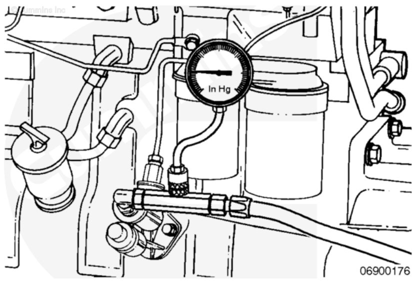
Check for a restriction between the fuel lift pump and the fuel injection pump.

1. Supply line to fuel filter
2. Fuel filter
3. Supply line to fuel injection pump
4. Fuel inlet line from tank.



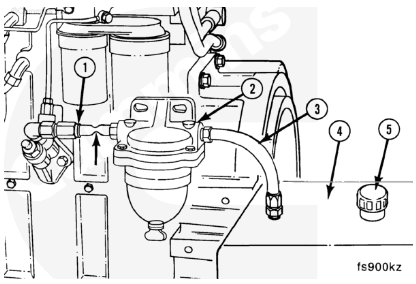
Measure the fuel lift pump inlet restriction with a vacuum gauge between the fuel lift pump inlet and the supply line (4) from the fuel tank.

kpa		in-hg
27	MAX	8

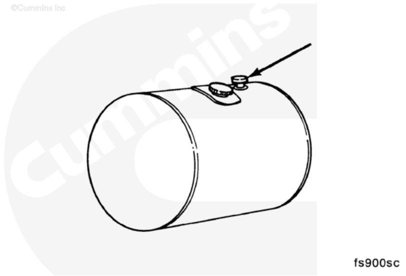


If the inlet restriction is above the maximum, check for restrictions or suction leaks in the fuel circuit to the fuel lift pump:

1. Supply line
2. Prefilter
3. Supply line
4. Supply tank
5. Tank vent.



Look for a plugged supply tank vent first.

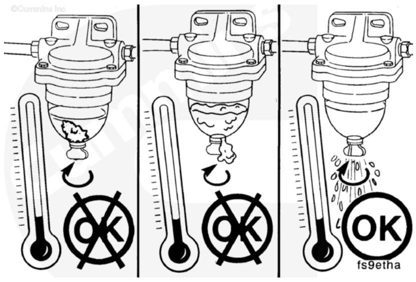


Fuel prefilters, inline and water separator type, can become clogged and cause a loss of fuel flow.

Check the prefilter for clogs or restrictions.

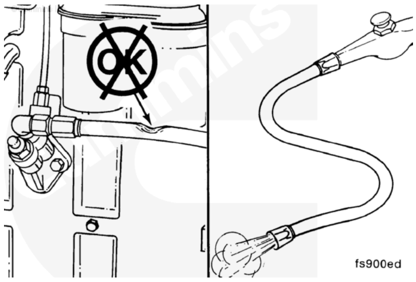
In cold weather, check the prefilter for gelled fuel.

Clean or replace the prefilter, if necessary.



Check for kinks or bends in the fuel supply line that can cause a restriction in the fuel flow.

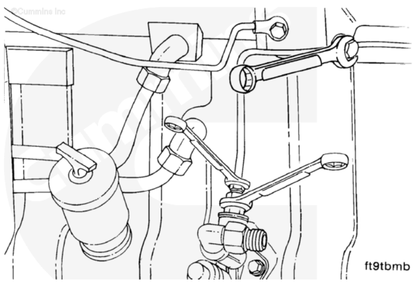
Remove and blow out the fuel supply lines.



Preparatory Steps

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

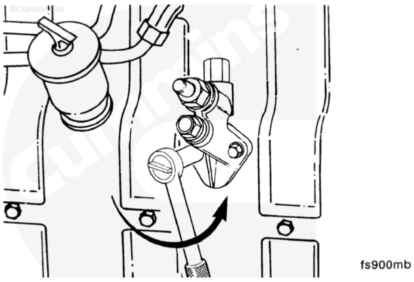


- Clean debris from the fuel line fittings and the fuel lift pump.
- Disconnect the low-pressure fuel lines. Refer to Procedure 006-024 (/qs3/pubsys2/xml/en/procedures/41/41-006-024.html).

Remove

Remove the two fuel lift pump mounting capscrews.

Remove the fuel lift pump.

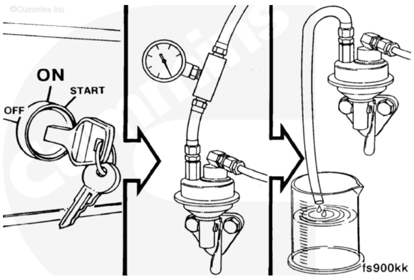


Test

The output of the fuel lift pump can be checked in two ways:

Test 1: Measure the output pressure using an in-line pressure gauge installed between the filter head and the fuel injection pump.

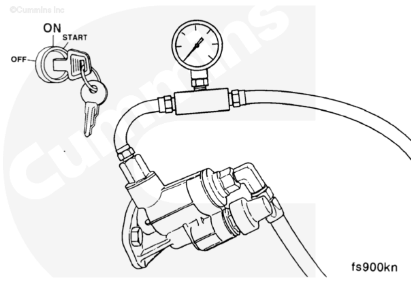
Test 2: Measure the flow volume.



Test 1: Output Pressure Test (Piston Style)

Operate the engine and measure the output pressure of the fuel lift pump using an in-line pressure gauge at the inlet to the injection pump.

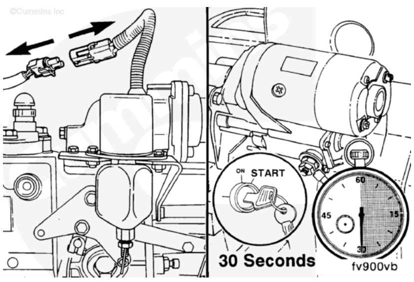
Minimum pressure at high idle is 138 kPa [20 psi].



Test 2: Flow Volume Test (Piston Style)

⚠ CAUTION ⚠

To prevent the engine from starting, disconnect the fuel shutdown wiring. Residual fuel in the injection pump can cause the engine to start.



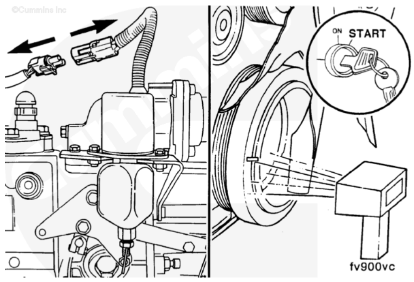
⚠ CAUTION ⚠

Do not crank the starter for more than 30 seconds at a time. Doing so can result in starter damage. Also, high voltage during cranking can damage the shutdown solenoid.

Disconnect the fuel shutdown solenoid wire.

Measure the engine cranking speed with a handheld tachometer, Part Number 3377462.

The minimum cranking speed is 120 rpm.



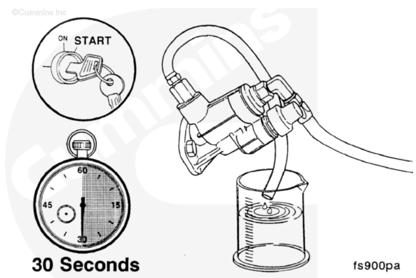
⚠ CAUTION ⚠

Leave the shutdown solenoid disconnected for the following check:

Disconnect the output pressure line from the fuel lift pump and run it into a container.

Crank the engine for 30 seconds and measure the fuel lift pump flow volume.

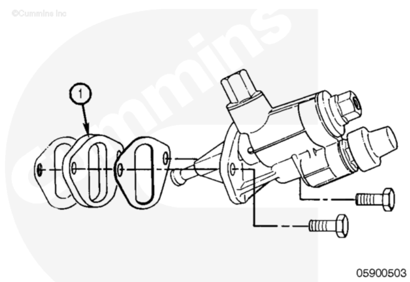
The minimum flow volume is 150 mL [5 oz].



Install

⚠ CAUTION ⚠

Alternately tighten the mounting capscrews. As the capscrews are tightened, the fuel lift pump plunger is pushed into the pump. Failure to tighten the capscrews in an even manner can result in the plunger being bent or broken, causing sticking and failure.



Piston Style

Install the pump.

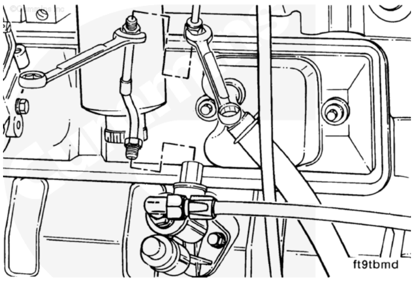
Torque Value: 24 n•m [18 ft-lb]

The 5-mm [0.20-in] spacer (1), Part Number 3914284, **must** be installed along with a gasket, Part Number 3931348, on each side of the spacer.

Note : For some applications, a bracket used for supporting other options will replace the 5-mm spacer.

Finishing Steps

- Install the fuel line to the fuel lift pump and fuel filter head. Refer to Procedure 006-024 (/qs3/pubsys2/xml/en/procedures/41/41-006-024.html).
- Vent the low-pressure fuel lines. Refer to Procedure 006-024 (/qs3/pubsys2/xml/en/procedures/41/41-006-024.html).



Last Modified: 13-May-2003